# BAHRAIN POLYTECHNIC

*MSc in Artificial Intelligence*

**TEMPLATE D**

# Demonstration & Viva Pack

**Submission of slides + plan: 1 day before scheduled demonstration**

*Demonstration window: Week 13*

| | |
|---|---|
| **Thesis Title** | CyberBoard AI: An Agentic Retrieval Augmented Advisory Board over Corporate Filings |
| **Student Name** | Hatem Isa |
| **Student ID** | 202508993 |
| **Primary Supervisor** | Dr. Joshua Samuel |
| **Second Marker** | [Second Marker] |
| **Scheduled Date & Time** | Tuesday 1 September 2026, [time] |
| **Room / Location** | [Room / Location] |

## Part 1: Demonstration Structure (45 minutes)

The demonstration is divided into three timed segments. Plan your content to fit within these limits; assessors will signal time and may cut presentation short if it overruns.

| Segment | Duration | Content |
|---|---|---|
| **A: Presentation** | 20 minutes | Slide-based overview of problem, methodology, results, contributions. |
| **B: Live Demonstration** | 15 minutes | Live demonstration of the implemented system with prepared scenarios. |
| **C: Viva Voce** | 10 minutes | Oral examination by supervisor and second marker. |

## Part 2: Slide Deck Plan (Segment A, 20 min)

*Build your slide deck to follow this exact structure. Aim for ~14 to 16 slides for a 20-minute presentation (about 75 to 90 seconds per slide). Title and references slides are not counted in time.*

| # | Slide | Suggested Time | Required Content |
|---|---|---|---|
| **1** | **Title** | 30 s | Title, your name, ID, supervisors, date, Bahrain Polytechnic logo. |
| **2** | **Agenda** | 30 s | Roadmap of the 20-minute presentation. |
| **3** | **Motivation** | 90 s | Why this problem matters now; a striking concrete example. |
| **4** | **Problem Statement** | 60 s | One sharp sentence; sub-points justifying scope. |
| **5** | **Research Questions** | 60 s | Bulleted RQs. |
| **6** | **Related Work** | 90 s | 3 to 4 closest prior works summarised; the gap you fill. |
| **7** | **Methodology Overview** | 120 s | Architecture diagram; high-level pipeline. |
| **8** | **Data & Pre-processing** | 60 s | Dataset(s), splits, key pre-processing steps. |
| **9** | **Proposed Approach** | 120 s | Your contribution; key technical novelty. |
| **10** | **Experimental Setup** | 60 s | Baselines, metrics, compute, statistical tests. |
| **11** | **Headline Results** | 120 s | Single most important results table/chart. |
| **12** | **Ablations & Errors** | 90 s | What helped, what didn't, where the system fails. |

| # | Slide | Suggested Time | Required Content |
|---|---|---|---|
| **13** | **Responsible AI** | 60 s | Bias, fairness, privacy, misuse, environment. |
| **14** | **Contributions Recap** | 60 s | 3 to 4 numbered contributions. |
| **15** | **Limitations & Future Work** | 60 s | Honest limitations and 2 to 3 directions. |
| **16** | **Q&A bridge / References** | 30 s | Thank-you slide; links to repo, paper, contact. |

### Slide design guidelines

- Use the Bahrain Polytechnic title slide template.
- Maximum 6 bullet points per slide; minimum 24 pt font. (suggestive only)
- Every figure must have a legend and be readable from a distance.
- Cite all external figures on the slide they appear (small footer text).
- Use animations sparingly and only to reveal information progressively.

## Part 3: Live Demonstration Plan (Segment B, 15 min)

*Prepare a sequence of 3 to 5 demonstration scenarios that together cover the system's core features, edge cases, and contribution. Have a recorded backup of each scenario in case of technical failure.*

| # | Scenario | What You Will Show | Expected Outcome | Time |
|---|---|---|---|---|
| 1 | Grounded board answer | Ask Agent mode. Ask: what are Apple's main risk factors in the most recent 10 K? Read the cited answer, the source badges, and the confidence score. | A correct, source attributed answer that ends with the advisory only disclaimer and adds no ungrounded claim. | 3 min |
| 2 | Retrieval reasoning trace | Search Filings mode. Query: cybersecurity risk factors. Open the trace: BM25 and dense candidates, 0.7 to 0.3 fusion, cross encoder rerank with logits and confidence. | A transparent pipeline. The reranker reorders the results and each carries a confidence score. | 3 min |
| 3 | Abstention and anti hallucination | Ask for a figure that is not in the corpus. Show that the agent declines rather than inventing a number. | No hallucination. The agent states that the evidence is not present and stops. | 3 min |
| 4 | Cross market, GCC | Ask a Bahrain Bourse question, for example | A grounded GCC answer drawn from | 3 min |

| # | Scenario | What You Will Show | Expected Outcome | Time |
|---|---|---|---|---|
| | | Alba's full year 2025 profit, to show the approach transfers to the second market. | the Bahrain Bourse corpus. | |
| 5 | Reproducible evaluation | Evaluation Results view. Show the FinanceBench hit rate of 96.8 percent and the ablation, rendered from stored result files. Name the repository commit of record, 9660fb8, so any figure can be verified against the exact code that produced it. | Measured performance is visible in the same tool, traceable to result files at a named commit. | 3 min |

### Pre-demonstration checklist

- ☐ Laptop fully charged plus charger; HDMI and USB-C adapters.
- ☐ System pre-installed and tested on the presentation laptop.
- ☐ All datasets and model weights downloaded locally (no live downloads).
- ☐ Internet connection tested; fall-back hotspot ready.
- ☐ Pre-recorded video of each scenario as backup (2 to 3 minutes each).
- ☐ Web UI / CLI / Jupyter notebook open at correct starting state.
- ☐ Terminal font size enlarged for visibility (16 pt+).
- ☐ Sensitive credentials redacted from screens and slides.
- ☐ Repository link printed/QR-coded on the title slide.
- ☐ Hard copy of slides and demo plan brought to the room.

## Part 4: Viva Voce Preparation (Segment C, 10 min)

The viva tests breadth, depth, and ownership of the work. Expect approximately 6 to 10 questions across the categories below. Prepare concise (≤90 second) answers in advance and rehearse out loud.

| Category | Representative Questions |
|---|---|
| **Motivation & Scope** | Why is this problem important? Why did you scope it this way? Who would use this? |
| **Methodology** | Why this model? Why this dataset? What alternatives did you reject and why? |
| **Implementation** | Walk us through the most complex 50 lines of your code. How would you scale it 100x? |

| Category | Representative Questions |
|---|---|
| **Evaluation** | Why these metrics? Were your statistical tests valid? What baselines did you miss? |
| **Results** | Why does your model fail on case X? What confounders might explain Y? |
| **Responsible AI** | What biases exist in your data and model? How could this system be misused? |
| **Originality** | What is your single biggest contribution? How is it different from FinSage [3] on retrieval and ReAct [7] on the agentic side? |
| **Reflection** | If you started again, what would you do differently? What surprised you most? |

## Tips for the viva

- Listen to the full question before answering. Paraphrase to confirm understanding.
- If you do not know an answer, say so honestly and reason out loud.
- Defend choices with evidence (a citation, an experiment, a measurement), not opinion.
- It is acceptable to disagree with a marker if you can justify your position.
- Have a printed copy of your key results table to refer to.
- Do not use generative AI tools during the viva. Phones, laptops not used for the demo must be put away.

### Part 4b Prepared Viva Answers (student preparation)

Concise answers to the representative questions in Part 4, each planned for under 90 seconds. These are preparation notes, not part of the assessed submission.

### Motivation and Scope

Board members must decide from filings that run to hundreds of pages, and general question answering tools give fluent but uncheckable answers. I scoped the system to evidence grounded, board level answers across two markets, the US and the GCC, because that combination is both useful and, as far as the review found, unstudied. The users are board members and governance staff who need a cited answer, not a chat.

### Methodology

I chose a ReAct agent over hybrid retrieval because the tools let the model read the exact passage a question needs, and I route to a Gemini flash model for cost and speed. The datasets are real filings, SEC EDGAR and Bahrain Bourse, with FinanceBench used for retrieval quality. I rejected multi agent routing as harder to evaluate and explain within a thirteen week thesis, and rejected a single dense retriever because it misses exact term matches such as a ticker or a line item name.

### Implementation

The most complex code is the hybrid retrieval and rerank path: BM25 and dense scores are normalised and fused at 0.7 to 0.3, then a cross encoder reranks the pooled candidates. This is also where the ticker filtering defect lived, filtering ran after the global top k, which I fixed by scoping candidates to the company first. To scale it 100x I would move the index to a managed vector store, batch and cache embeddings, and parallelise the judge; retrieval is already sub second.

### Evaluation

I report accuracy, faithfulness, hallucination rate, and retrieval hit rate. Because the agent and the baselines answer identical questions, the paired McNemar test is the correct significance test; it is valid but underpowered at 35 questions, which is why I report a five run mean rather than a single result. The baselines I would add next are a stronger single pass RAG with reranking and a larger hand verified set.

### Results

The agent fails on one compound GCC question that asks for two quantities at once, the owner equity and the earnings per share of the KFH Bahrain entity, and it occasionally returns only one; this is answer scoping, not retrieval. A confounder to note is that closed book accuracy is far higher on US mega caps because their figures sit in the model's parametric memory, which is exactly why the GCC set is the honest test of retrieval.

### Responsible AI

The main data bias is coverage: large US companies are over represented and small GCC issuers under represented, so the system is strongest where filings are richest. The clearest misuse would be treating an advisory output as a decision, so every answer carries an advisory only disclaimer and the agent abstains when no evidence is retrieved rather than guessing.

### Originality

My single biggest contribution is the assembled and evaluated whole: hybrid retrieval, an agentic reasoning layer, and answer level explainability for corporate governance, measured on two markets with a reproducible harness. It differs from FinSage, which stops at retrieval and is not tested on GCC, and from ReAct, which is general purpose and not grounded in a filing corpus.

### Reflection

If I started again I would build the faithfulness judge with the full retrieved context from day one, because a truncated judge produced a misleading 67 percent faithfulness early on that was a measurement artefact, not a property of the agent. What surprised me most was the honest ablation: the cross encoder reranker helped on the larger US corpus but slightly hurt recall on the small, homogeneous GCC set.

## Part 5: Assessor Scoresheet (for marker use)

*This sheet is used by the supervisor and second marker. Students should be aware of the criteria but not complete this section.*

| Criterion | Weight | Mark /100 | Comments |
|---|---|---|---|
| **Presentation Quality** | **20%** | | |
| **Technical Depth of Demonstration** | **25%** | | |
| **Justification of Design & Methodology** | **20%** | | |
| **Response to Viva Questions** | **20%** | | |
| **Reflection & Future Work** | **10%** | | |
| **Professional Conduct & Materials** | **5%** | | |
| **OVERALL WEIGHTED MARK** | **100%** | | |

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